Layal Canoe

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Portfolio | GitHub

Education

Effat University — Jeddah, Saudi Arabia
Bachelor of Computer Science – Artificial Intelligence — *2022 – 2026*

- CGPA: 3.97/4.00
- 50% Merit Scholarship; Dean's List 2022–2023 , 2023–2024 and 2024–2025

Experience

AI Intern — Jun. 2026 – Present
Heroz — *Jeddah, Saudi Arabia*

- Built a hybrid recommendation system that ranks personalized children's activities using interests, preferences, developmental goals, behavior, exploration, and session suitability.
- Designed the MongoDB data architecture and Neo4j Knowledge Graph, connecting children, activities, interests, goals, learning outcomes, and behavioral data.
- Developed graph-building and multi-hop traversal logic using JavaScript, Node.js, Neo4j, and Cypher to generate and retrieve recommendation candidates.
- Built AI automation workflows with n8n, integrating MongoDB, AI-generated personalized content, WhatsApp delivery, and database updates for automated outreach.

Back-End Developer Intern — Summer 2025
Breifiction – Marketing Agency — *Jeddah, Saudi Arabia*

- Built 5+ REST API endpoints powering a personalized AI assistant for customer messaging and automated shipment-tracking and order-status responses.
- Integrated AI functionality with client databases and designed SQL/NoSQL schemas supporting real-time, context-aware data retrieval and AI workflows.
- Implemented authentication and backend security practices for production-oriented services.

Research & Publications

Scalable Multi-Agent Edge Intelligence for Traffic Coordination | *NLP & Big Data*

- Built a distributed AI pipeline for multi-agent traffic coordination across 8 Jeddah intersections using Kafka, Hadoop HDFS, Spark/PySpark, and Spark MLlib, benchmarking 1,000,000 sensor records with an 80.6× throughput improvement.
- Developed a severity classifier achieving 100% accuracy on clean data and 84.9% under 20% adversarial sensor dropout.
- Integrated Vehicle, Incident Detection, and Traffic Light agents with RAG and Knowledge Graph reasoning for real-time traffic decision-making.

IEEE Conference Publication | *AI & Cybersecurity in Powerline Communication Systems*

- Peer-reviewed research accepted at the IEEE SCC Conference in Sousse, Tunisia and selected for research funding by Effat University's Deanship of Graduate Studies and Research.
- Investigated cybersecurity challenges in AI-enabled Powerline Communication systems for smart-grid and IoT infrastructure, including AI-driven anomaly detection and post-quantum cryptography.

Selected Projects

AgentNation – AI Policy Simulator | *Python, Pandas, NumPy, Multi-Agent Systems*

- Designed a four-layer multi-agent architecture for cross-ministry policy simulation and built a functional prototype linking policy-impact and workforce-intelligence services.
- Modeled workforce supply-demand gaps and synthesized 22 sources to identify 7 governance capability gaps that informed system requirements.

Hajj Buddy AI | *RAG, NLP, LLMs, Node.js*

- Built a Retrieval-Augmented Generation assistant for Hajj and Umrah pilgrims using verified Islamic sources to reduce misinformation in generative AI responses.
- Designed a grounded retrieval pipeline and deployed a live conversational interface focused on accuracy, trust, and usability.

IoT Intrusion Detection System | *Python, Machine Learning, Deep Learning*

- Developed and evaluated an ML pipeline on approximately 200,000 CIC IoT 2023 network samples using SMOTE and PCA.
- Benchmarked Random Forest, XGBoost, MLP, and CNN models, achieving highest accuracy of 95% with Random Forest.

Music Map – Full-Stack Music Discovery Platform | *JavaScript, PHP, SQL, HTML/CSS*

- Developed a full-stack music discovery platform with an interactive frontend and PHP/SQL backend supporting CRUD operations, authentication, and database-driven content management.

dormOra – AI Dermatology Assistant | *Computer Vision, RAG, Knowledge Graphs*

- Selected as part of Effat University's sole representing team in the BMC 2026 national hackathon.
- Built an end-to-end AI prototype analyzing skin images across acne, pigmentation, redness, and texture using RAG and Knowledge Graph reasoning.

Technical Skills

Languages: Python, JavaScript, Java, C++, SQL, PHP, Cypher, HTML/CSS
AI & Machine Learning: Machine Learning, Deep Learning, RAG, LLM Integration, NLP, AI Agents, Multi-Agent Systems, Knowledge Graphs, Computer Vision
Data & Big Data: MongoDB, Neo4j, Apache Spark/PySpark, Kafka, Hadoop HDFS, Spark MLlib, Pandas, NumPy
Backend & Automation: Node.js, REST APIs, SQL/NoSQL Databases, n8n, Workflow Automation
Developer Tools: Git/GitHub, Docker, Linux, Postman, Streamlit

Honors & Awards

- DGRS Research Funding Award – Effat University
- KAUST Academy AI Specialization – Stage 2 Selection, National Competitive Assessment, 2025